

Experiment 1 : Comprehensive EDA and Project Documentation

Total Marks: 100

1. Learning Objectives

Upon successful completion of this assignment, students will be able to:

- Structure and write a formal Exploratory Data Analysis (EDA) report.

- Articulate the *theory* behind data types, distributions, and feature engineering.

- Move beyond single questions to a holistic analysis of a dataset.

- Master the use of `matplotlib` and `seaborn` for creating insightful visualizations.

- Analyze and interpret the distribution of a target variable (univariate analysis).

- Analyze and interpret the relationship between features and the target (bivariate analysis).

- Justify data cleaning and feature engineering decisions in a professional context.

2. Introduction

In Lab 1, you "answered" specific questions about the datasets. This is a good first step. However, a data scientist's primary job is not just to answer questions, but to *tell the story* of the data, identify insights, and prepare the data for machine learning.

This assignment requires you to take your work from Lab 1 (specifically, **Assignment 1: The Flight Price Dataset**) and create a complete, high-quality EDA report. You will revisit the questions from Lab 1, but this time as *starting points* for a deeper investigation.

Your deliverable is a single, clean report (in PDF format) that reads like a professional analysis. It must include all your markdown explanations, the code you used, the code's output, and all visualizations. **The quality of your explanations and interpretations is as important as the code itself.**

3. Prerequisites

Ensure your Python environment has the following libraries installed:

```
pip install numpy pandas matplotlib seaborn
```

4. Experiment Tasks

You must use the **Flight Price Dataset** (from Lab 1, Assignment 1).

Task 1: Project Introduction & Data Overview (15 Marks)

This section sets the stage for your report.

1. **Project Goal:** Write a 1-2 paragraph introduction. What is the dataset? What is the business problem we are trying to solve? (e.g., "The goal of this analysis is to understand the key factors that drive flight ticket prices...").
2. **Data Loading:** Include the code to load the dataset.
3. **Initial Inspection:** Display the `shape`, `info()`, and `describe()` outputs.
4. **Data Dictionary & Theory:**

Re-create the **MetaData** section from Lab 1.

After the dictionary, write a short explanation of the **theoretical difference between numerical and categorical data**. Classify each feature from the dataset (e.g., 'Airline', 'Source') into one of these two types.

Task 2: Data Cleaning & Preprocessing (25 Marks)

This section details how you identified and fixed data quality issues.

1. **Missing Values:** Include code to identify and report the number of missing values in each column.
2. **Data Type Conversion ('Duration'):**

Show the "before" state (e.g., `df["Duration"].head()`).

Provide the code you used to convert 'Duration' into a numerical value (e.g., total seconds, as in Lab 1, Q5).

Justification (Theory): Explain *why* this step is necessary. Why can't a machine learning model use "2h 45m"? What is the advantage of using a single numerical unit like seconds or minutes?

3. **String Inconsistency ('Additional_Info'):**

Show how you identified the 'No Info' vs. 'No info' problem (e.g., `value_counts()`).

Provide the code to standardize this (as in Lab 1, Q4).

Justification: Explain why this string inconsistency is a problem for analysis.

Task 3: Target Variable Analysis ('Price') (20 Marks)

A deep analysis always begins with the target variable.

1. **Statistical Summary:** Show the `describe()` output for the 'Price' column.

2. **Visualization (Distribution):**

Plot a **histogram** of the 'Price' column.

Plot a **boxplot** of the 'Price' column.

3. **Interpretation:**

Based on the histogram, describe the distribution. Is it normal, left-skewed, or right skewed?

Based on the boxplot, what is the approximate median price? Are there many outliers? What might these outliers represent in the real world?

Task 4: Exploratory Data Analysis (EDA) (30 Marks)

This is the core of your report, where you explore relationships in the data.

1. **Univariate Analysis (Categorical):**

Create **bar charts** for 'Airline', 'Source', and 'Total_Stops'.

Insight: Write 1-2 sentences for each chart. (e.g., "Which airline has the most flights in this dataset?", "What is the most common number of stops?").

This covers Lab 1,
Q2.

2. **Bivariate Analysis (Feature vs. Target):** This is the most important part of EDA. **Categorical vs. Price:**

Create **boxplots** for 'Airline' vs. 'Price'.

Create **boxplots** for 'Total_Stops' vs. 'Price'.

Create **boxplots** for 'WeekDay' vs. 'Price'. (This covers Lab 1, Q3).

Numerical vs. Price:

Create a **scatter plot** for 'Duration_in_seconds' vs. 'Price'.

3. **Interpretation:** For *each* bivariate chart, write a 2-3 sentence insight.

(e.g., "The boxplot clearly shows that 'Jet Airways' has a higher median price than 'IndiGo'.")

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(e.g., "As seen in the boxplot, there is a strong positive relationship between the number of stops and the price.")

(e.g., "The scatter plot shows a positive correlation: flights with longer durations tend to be more expensive.")

Task 5: Feature Engineering & Final Conclusion (10 Marks)

This section discusses the creation of new features and summarizes your findings.

1. Feature Engineering:

Show the code you used to engineer the 'Dep_Time' and 'Arrival_Time' columns into categories like "Morning", "Afternoon", etc. (from Lab 1).

Justification (Theory): Explain *why* this is a useful feature. Why is "Morning" a better feature than the raw time "08:15"? (Hint: It groups similar times, reducing noise and capturing time-of-day pricing).

Create a new bar chart showing the average 'Price' by your new time-of-day category (e.g., 'Dep_Time_Category').

2. **Summary of Insights:** Write a final conclusion (3-5 bullet points) summarizing the most important insights from your entire analysis.

(e.g., "Insight 1: 'Total_Stops' is a major driver of price.")

(e.g., "Insight 2: 'Jet Airways' is consistently the most expensive airline.")

5. Submission Guidelines

Submit a single **.zip** archive containing:

1. PDF Report (**StudentID_Report.pdf**):

This is the **primary deliverable**.

You must export your final Jupyter Notebook to a PDF. (In Colab/Jupyter, you can

do this via **File > Print > Save as PDF**).

The PDF must contain **all** your content: the markdown explanations, the code cells, the code outputs, and all visualizations.

Your report must be well-structured, with clear headings for each task, and be easy to read from top to bottom.

2. Source Code (.ipynb):

Include the original Jupyter Notebook (.ipynb) file as a backup.

Report Standards:

Every code cell must be followed by its output in the PDF.

Every visualization must be followed by a clear explanation (in a markdown cell, also visible in the PDF) detailing the insight you gained from it.

The report should be clean, professional, and tell a complete story of your analysis.

